

Big Data Computing (DTS 302)

A Reproducible Analysis of Nigerian Spotify Trends using Jupyter

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Abstract

This report presents a comprehensive, reproducible analysis of the "nigerian_spotify_songs" dataset conducted within the Jupyter Notebook framework. Our team focused on leveraging the notebook's literate programming capabilities to document a full data science workflow: from importing libraries and loading data, to cleaning, visualizing, and interpreting results using SHAP (SHapley Additive exPlanations). Our findings highlight the significant impact of a track's release year, its primary genre, and its danceability on its popularity score, offering data-driven insights for the Nigerian music industry. The notebook is structured to run sequentially without errors, ensuring full reproducibility.

1. Introduction

This document serves as the final report for the DTS 302 lab assessment. The goal was to use the Jupyter Notebook environment to conduct a thorough analysis of a dataset containing audio features and metadata for 530 Nigerian tracks on Spotify. The analysis focused on understanding the factors that correlate with a song's popularity.

1.1 Jupyter Notebook as an Analytical Tool

Jupyter Notebook was chosen for its unique suitability for data analysis tasks.

- **What is Jupyter Notebook?** It is an open-source, web-based interactive computing environment that allows users to create and share documents containing live code, equations, visualizations, and narrative text.
- **Key Feature 1: Incremental Computation and Rapid Prototyping.** The cell-based execution model allows analysts to write, execute, and debug code in small, logical

sections. This removes the traditional "edit-compile-run" cycle, accelerating the process of exploring data and testing hypotheses.

- **Key Feature 2: Integrated Documentation.** By combining code with rich Markdown cells, the entire analytical narrative—including the rationale for each step, interpretation of results, and final conclusions—is preserved within a single, self-contained file. This is invaluable for peer review, collaboration, and ensuring the long-term reproducibility of the work.

2. Analytical Workflow

2.1 Data Preparation

The analysis began by importing necessary libraries: `Pandas`, `NumPy`, `Matplotlib`, `Scikit-learn`, and `shap`. The dataset was loaded, and an initial inspection was performed.

2.2 Handling Data Anomalies

During the exploration phase, we discovered that the `artist_top_genre` column contained 53 instances of the string `'Missing'`, indicating an absent genre label. Simply dropping these rows would have resulted in a 10% loss of data and could bias the analysis. We therefore opted to recode these entries as a new category, `'Unknown'`. This method preserves all other track data (like danceability, energy, etc.) for use in our numerical analyses and SHAP modeling.

2.3 Visual Analysis and Interpretation

To address our objective of identifying patterns, we generated several plots:

- A **line plot** was used to visualize the trend of average track popularity over different release years.
- A **bar chart** was created to compare average popularity across different musical genres.
- A series of **box plots** provided a visual summary of the distribution and outliers for all continuous numerical features (e.g., loudness, tempo, acousticness).

2.4 Predictive Modeling and Explanation (SHAP)

To move beyond simple correlation and understand the directional impact of each feature, we trained a Random Forest Regressor to predict track popularity. We then applied SHAP (SHapley Additive exPlanations). SHAP values provide a unified measure of feature importance, showing not only which features are most important, but also whether high or low values of that feature push the model's prediction higher or lower.

3. Findings and Strategic Insights

3.1 Popularity Trends Over Time

The line plot reveals a significant pattern: average popularity rises sharply from 2008, peaking around 2015-2016. This aligns perfectly with the documented international rise of Afrobeats artists. The subsequent sharp decline is a known statistical artifact of Spotify's algorithm, which gives more weight to recent streams; hence, older tracks have an inherent advantage in cumulative popularity scores.

3.2 Genre Performance

The bar chart demonstrates a clear hierarchy in streaming appeal. Afropop and Afro Dancehall are the top-performing genres, while more niche styles like Alternative R&B and Nigerian Hip Hop lag significantly behind. This suggests that for artists aiming for high Spotify visibility, aligning with these mainstream genres is a more effective strategy.

3.3 Audio Feature Distribution

The box plots confirm the core characteristics of the dataset. The tight distribution and high median of **danceability** confirm the dance-centric nature of the music. The extreme right-skewness of **instrumentalness** and **acousticness** shows that instrumental and acoustic tracks are rare outliers; the market overwhelmingly prefers vocal-led, electronically produced songs.

3.4 SHAP-Driven Feature Importance

The SHAP analysis provided the most nuanced insight. The top three most impactful features for predicting popularity are:

1. **Release Date (Year):** The model confirms the strong negative relationship between recency and predicted popularity.
2. **Artist Top Genre:** Belonging to a mainstream genre like Afropop provides a strong positive boost, while niche genres negatively impact the prediction.
3. **Danceability:** Higher danceability scores consistently and positively influence the model's output.

The analysis also showed that high **loudness** (less negative dB) and low **instrumentalness** are also strong positive indicators of popularity.

4. Conclusion

Through a systematic, reproducible workflow in Jupyter Notebook, we have successfully analyzed the Nigerian Spotify Songs dataset. Our key observations confirm the dominance of Afropop and Afro Dancehall genres, the time-based bias in popularity scores, and the overwhelming market preference for danceable, vocal-driven, and electronically produced tracks. The SHAP analysis was particularly powerful, quantifying the directional impact of each feature. This report and the accompanying notebook provide a clear, verifiable template for data-driven decision-making in the music streaming industry.